Rayen Chemlali

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SUMMARY

Currently a student at INP Toulouse – ENSEEIHT (N7) on an academic exchange, I am completing the final year of my engineering degree, majoring in HPC & Big Data, alongside a Master's degree in the PSMSC track (Scientific Computing). I come from a 4-year Software Engineering program at INSAT (National Institute of Applied Science and Technology) in Tunisia. Passionate about mathematics and computer science, I have taken part over the course of my studies in several programming contests and hackathons, alongside gaining professional experience through internships, freelance work and part-time roles.

KEY SKILLS

- **Generative & Agentic AI:** NLP, LLMs (prompt engineering, fine-tuning, embeddings), RAG, Agentic AI, multi-agent architecture, tool calling, LangChain, LangGraph, Qdrant
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, CNN, Transformer Architecture, time series, audio classification
- **Scientific Computing & HPC:** parallel and distributed computing, HPC algorithms, OpenMP, MPI, CUDA, OpenCL
- **Optimization:** Operations research (LP, MILP), heuristic search algorithms (A*), Gurobi
- **Big Data:** Hadoop, Spark, Kafka
- **Programming Languages:** Python, C++, Java, Go, JavaScript, PHP
- **Frameworks:** FastAPI, NestJS, ASP.NET, Symfony, React, Next.js, Angular
- **Software Development & Tools:** Software architecture, Design Patterns, Relational and non-relational databases, Git, Docker, Kubernetes, CI/CD, DevOps

PROFESSIONAL EXPERIENCE

Plug & Tel

ML Intern *Jun 2026 – Aug 2026*

- Reduced the classification error rate of a production AI model for answering machine detection (AMD).
- Kept latency compatible with the real-time constraints of VoIP streams.
- Explored several approaches: tabular models, CNNs on mel-spectrograms and hybrid models combining both.
- Implemented model explainability: Grad-CAM for CNNs and tabular model explainability.

Part-Time Software Engineer *Nov 2025 – Apr 2026*

- Developed VoIP microservices in Go.
- Took part in the production deployment process.

Elevvo Pathways

ML Intern *Mar 2026*

- Completed 5 end-to-end ML projects: clustering, classification, time series forecasting, computer vision and audio classification.
- Mastered the full ML pipeline, from data preprocessing to model evaluation. *Code*

Talan Tunisie Consulting

Summer Intern *Jun 2025 – Aug 2025*

- Designed and developed the first phase of the Smart Container Registry to optimize Docker storage costs.
- Conversational chatbot with dynamic tool calling.
- Unified Kubernetes/S3 dashboard and custom deletion rules. *Frontend* *Backend* *Demo*

Freelance

Software Developer *Various dates*

Worked on two freelance projects:

- **Strategy Copilot**: AI-powered business strategy copilot with a chatbot and graphical visualizations, generating diagnostics, financial projections and market analyses with PDF exports.
- **Shannon**: Co-developed a C++ code generator for electronic boards for the company Shannon, based on graph modeling, with a PyQt5 desktop application and an interactive Simulink-inspired canvas. *Demo*

PROJECTS

- **Multi-Agent Travel Planning System** *LLMs, LangChain, LangGraph, RAG, Qdrant* Jan 2026 – Jun 2026
  - End-of-year project: personalized travel planning driven by multiple AI agents, with conversational interaction and RAG.
  - Progressive knowledge base construction and multi-level persistent memory.
  - Optimized response time through background pre-computation.
  - Evaluated with automated tests simulating users with different personas. *Demo* *Report (full architecture details)*
- **Smart City Platform** *Hadoop, Spark, Kafka, MLlib* Jan 2026 – Jun 2026
  - Urban intelligence platform with a full Big Data pipeline: weekly batch and real-time streaming on 6 real-world datasets (traffic, energy, crime).
  - Computes via MLlib the weights of the City Pulse Score, a composite indicator per district for continuous urban monitoring.
- **Respiratory Sound Classification (ICBHI 2017)** *Python, Audio Transformers, CNN, BiLSTM* Feb – Apr 2026
  - Deep Learning research project on detecting abnormal respiratory sounds.
  - Reproduced a recent paper, then tested several approaches to improve it: CNN-BiLSTM with attention, DeiT on Mel and cochleagram, AST with SAM, Focal Loss, logit adjustment.
  - Reached 73% sensitivity versus 68% for the paper. *Code* *Report*
- **SoundGuard (Samsung Innovation Campus)** Oct 2025 – Nov 2025
  - AI program capstone project: real-time detection of dangerous noise levels to protect workers' hearing.
  - Classifies audio signals and triggers alerts when exposure exceeds safety thresholds. *Demo* *Code*
- **MediSys** *NestJS, Next.js, GraphQL, Python, LLMs, OCR* Mar 2025 – Jun 2025
  - Medical management platform: consultations, appointments, prescriptions and lab tests.
  - Automatic digitization of medical documents via OCR.
  - Conversational chatbot that recommends nearby doctors and medications tailored to the patient's case. *Frontend* *Backend* *Demo*
- **Heterogeneous GPU Computing (CUDA/OpenCL)** *CUDA, OpenCL, cuBLAS, C++* Apr – May 2025
  - Two projects completed within the same course.
  - Dynamic load balancing across CPU, iGPU and dGPU in OpenCL: 1.23× speedup over the GPU alone.
  - CUDA kernel optimization (tiling, float4 vectorization, TensorCores): 1235 GFLOPS, within 6% of cuBLAS performance. *Reports*
- **Operations Research Optimization** *Python, Gurobi, Next.js, FastAPI* Apr – May 2025
  - Modeling and solving three NP-hard problems with Gurobi (transportation, VRP with MTZ subtour elimination, facility location).
  - Dual and sensitivity analysis.
  - Two complete applications: a Next.js/FastAPI web platform with interactive route visualization, and a PyQt5 desktop application. *Code* *Report*

EDUCATION

INP Toulouse – ENSEEIHT *2026 – 2027*
– 5th year, HPC Big Data track (academic exchange)
– Master's degree, PSMSC track (Scientific Computing)

National Institute of Applied Science and Technology (INSAT) *2022 – 2026*
1st to 4th year in Software Engineering

Baccalaureate in Mathematics, Tunisia *2022*
Highest Honors

LANGUAGES

Arabic – Native **French** – Fluent **English** – Fluent **Spanish** – Beginner

CERTIFICATIONS

- Samsung Innovation Campus AI Program
- Efficient LLM Customization (NVIDIA)
- Building RAG Agents with LLMs (NVIDIA)

ACHIEVEMENTS

- Thematic winner (Enedis track) – AI For Impact Hackathon by La Mêlée, Oct 2026
- Top 6 – Data Quest 2026, Feb 2026 *Code* *Demo*
- 3rd place – AI 4 New Generation Hackathon, EY & Univ. Ibn Khaldoun, 2024
- Bronze medal (11th) – TCPC 2024
- 2nd place – Code Quest 2023, INSAT